

Experiment 2

Question

Perform basic image handling and processing operations on an image. Read an image in Python and convert the image to a Blur using Gaussian Blur.

Aim

To read an image using Python and OpenCV and apply Gaussian Blur to smooth the image.

Procedure

1. Import the OpenCV (cv2) library.
2. Read the input image using cv2.imread().
3. Check whether the image is loaded successfully.
4. Display the original image.
5. Apply Gaussian Blur using the cv2.GaussianBlur() function.
6. Display the blurred image.
7. Save the blurred image.

Code

```
import cv2

# Read image
image = cv2.imread("house.jpg")

# Check whether image is loaded
if image is None:
    print("Image not found!")
else:
    # Display original image
    cv2.imshow("Original Image", image)

    # Apply Gaussian Blur
    blur = cv2.GaussianBlur(image, (15, 15), 0)

    # Display blurred image
    cv2.imshow("Blurred Image", blur)

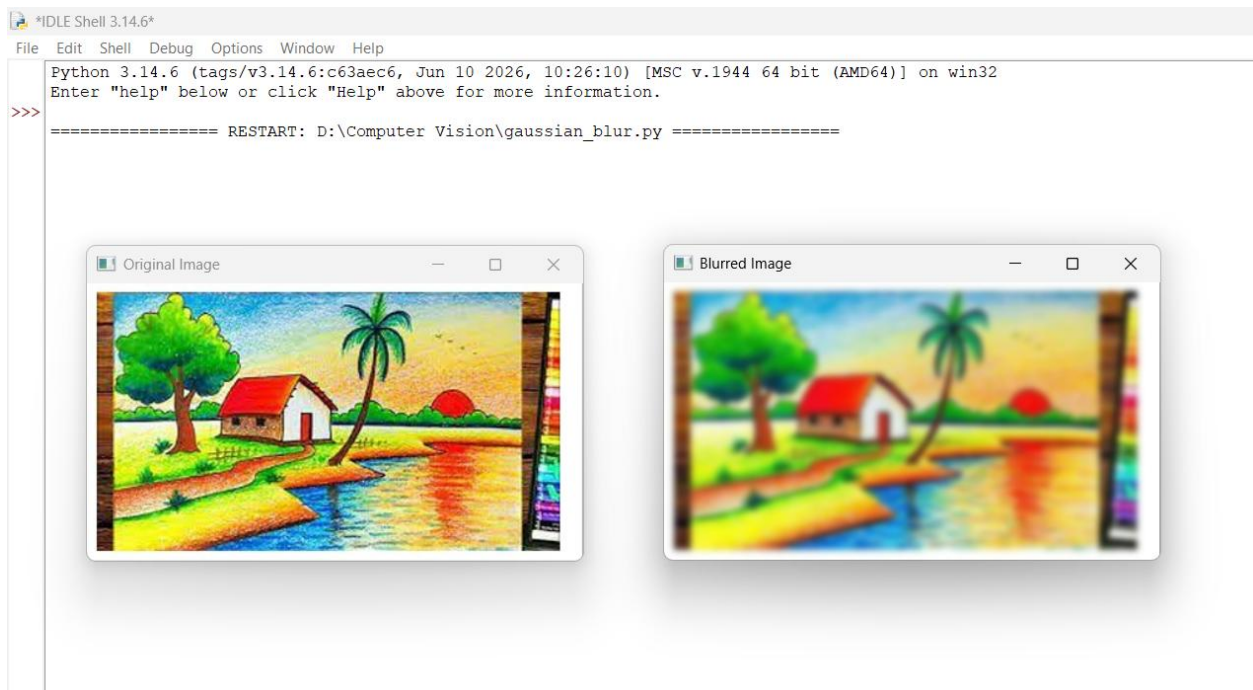
    # Save blurred image
    cv2.imwrite("blur_house.jpg", blur)
```

Input

Input Image: house.jpg

Output

1. **Original Image** : Displays the original color image.
2. **Blurred Image** : Displays the image after applying Gaussian Blur.
3. **Saved File** : blur_house.jpg is created in project folder.



Result

The input image was successfully read using OpenCV, Gaussian Blur was applied to smooth the image, and the blurred image was displayed and saved successfully.